DIRECTORY

By the turn of the 19th century, science had become intimately linked with progress. As the 20th century unfolded, cutting-edge research fueled crucial shifts in fundamental ideas. Discoveries abounded and rules were rewritten at both subatomic and cosmological levels.

WILHELM CONRAD RONTGEN

1845–1923

German physicist Wilhelm Röntgen made one of the most important advances in physics and medicine when he discovered X-rays in 1895. He found that electrically charged vacuum tubes emitted rays that made a fluorescent screen glow. These electromagnetic rays went though human skin to expose photographic plates but were blocked by metal and bone. Although the discovery earned him the first Nobel Prize in Physics in 1901, he bequeathed the prize money to scientific research and never patented the X-ray. Röntgen is also known for his discoveries in mechanics, heat, and electricity.

IVAN PAVLOV

1849–1936

Russian-born Ivan Pavlov abandoned a religious career to become a professor at the Military Medical Academy in St. Petersburg in 1890. He was director of the physiology department at the Institute of Experimental Medicine when he began researching the digestive secretions of dogs. He found that the dogs learned to associate the arrival of food with the sound of a bell. After a while, even when no food was provided, the dogs still salivated in response to the bell being rung. This is now called classical, or Pavlovian, conditioning. In 1904, Pavlov won the Nobel Prize in Physiology or Medicine for his work.

KITASATO SHIBASABURO

1853–1931

Japanese physician and bacteriologist Kitasato Shibasaburō studied in Tokyo and Berlin and developed serum therapy to protect against tetanus and diphtheria. In 1890, he discovered that injections of his tetanus serum, which contained the antitoxin that had been produced in the blood of an animal exposed to the tetanus bacteria, conferred immunity on the animal to which it was given. He went on to apply the same principle to protect against diphtheria.

JULES HENRI POINCARE

1854–1912

Mathematical physicist Henri Poincaré was born in Nancy, France. While trying to prove that the solar system is stable, he noticed that even tiny changes in the initial conditions of a system often result in large—and unpredictable—changes in outcome; in other words, his results exhibited chaotic behavior. Published as early as 1908, his findings about chaos were initially overlooked, but several decades later, they became the foundation for chaos theory. He also wrote papers on electromagnetism that informed Einstein's work on relativity.

J. J. THOMSON

1856–1940

English physicist J. J. Thomson was one of the first scientists to describe the structure of atoms. He identified "corpuscles," later called electrons, using a cathode ray tube. The particles had a negative electric charge and were about 2,000 times lighter than a hydrogen atom. In his plum pudding model, Thomson suggested that every atom consists of a number of electrons and an amount of positive charge to balance their negative charges. He thought the positive charge was spread throughout the atom and the electrons existed within it, like plums in a plum pudding. His discovery revolutionized the theories of atoms and electricity. He also confirmed the existence of isotopes—elements that each have several types of atoms, chemically identical but differing in weight.

SVANTE ARRHENIUS

1859–1927

After studying physics at the University of Uppsala in his native Sweden, Svante Arrhenius became professor of physics